

Forecasting & Predictive Analytics Guide

Methodology Behind Growth Forecasting and AI Conversational Summaries

1. Trend-Based Revenue Forecasting

CIVORA calculates next-period revenue forecasts based on historical sales velocity.

Formula: $\text{Next Revenue} = \text{Current Revenue} * (1 + (\text{Growth Rate} / 100))$.

Fallback logic: If growth rate cannot be calculated (e.g., first month of data), a default baseline growth rate of 8.0% (0.08 multiplier) is applied.

Best Practice: Upload at least three consecutive months of data for reliable trend mapping.

2. AI Chatbot Architecture

CIVORA features an AI panel powered by Watsonx Orchestrate, with a local rule-based fallback system.

The AI reviews dashboard context (aggregate statistics, regional trends, category mix, and year filters) to answer user questions.

3. Local Conversational Fallback Engine

If the AI server is offline, the local fallback processes user text using keyword matching:

- * 'report' / 'summary' queries: Returns a text report highlighting consolidated revenue, profit, units sold, and the latest period's revenue.
- * 'compare' / 'vs' queries: Calculates the growth rate between the first and last uploaded months and provides growth metrics.
- * General queries: Provides system help, prompting users on how to ask about metrics, charts, or periods.